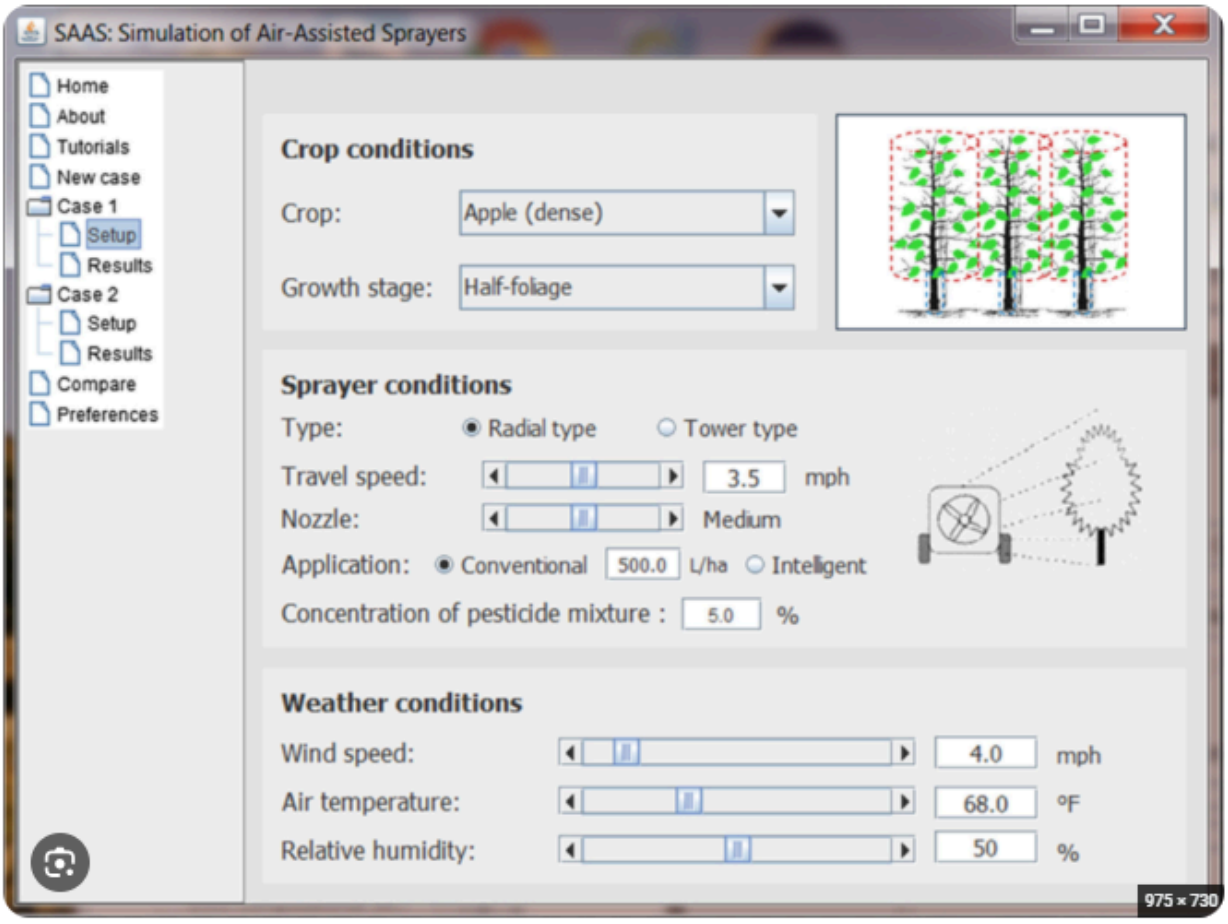
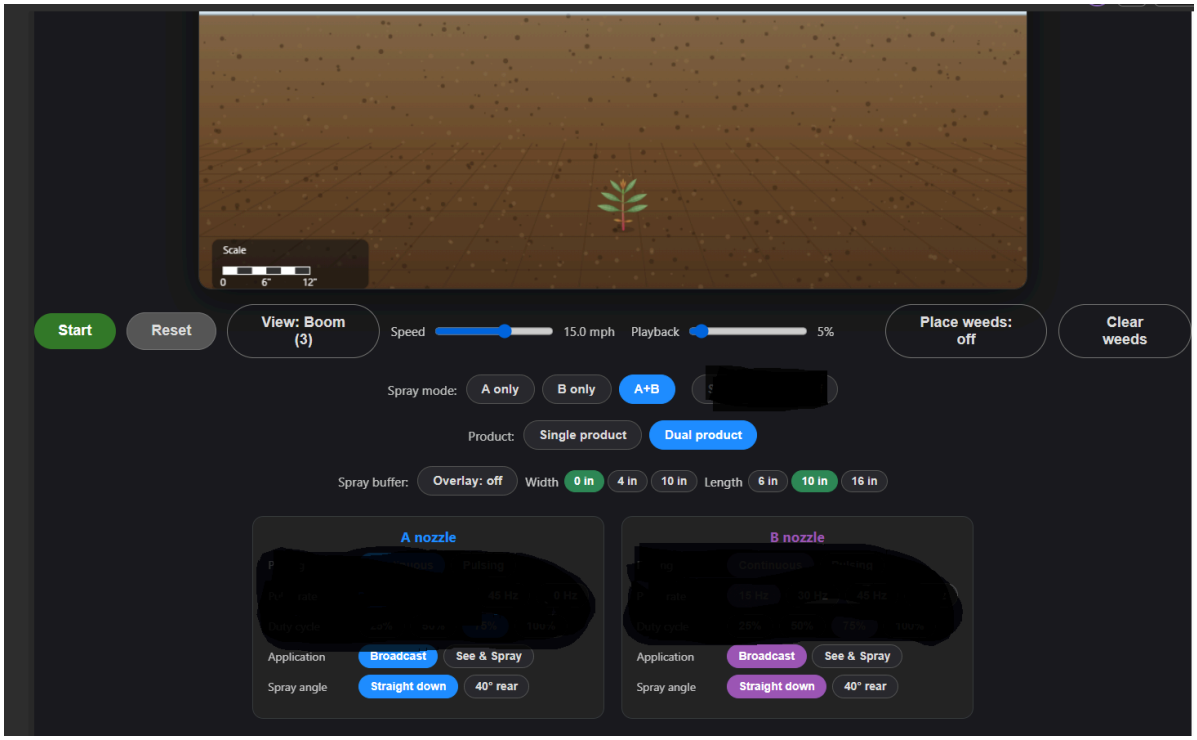
CoverageAI Technical Proposal: Closed-Loop Spray Simulation Application

CoverageAI is an intelligent system engineered to optimize agricultural spray deposition and coverage efficiency through an end-to-end, closed-loop feedback mechanism. The system dynamically evaluates environmental conditions and continuously modulates machine operating parameters to maximize canopy penetration and uniformity while minimizing drift. To demonstrate this architecture prior to field deployment, a high-fidelity simulation application will model real-time sensor inputs—including ambient environmental data, dynamic sprayer settings, and empirical spray validation metrics—to iteratively refine nozzle flow and pressure outputs. Furthermore, the simulation could ingest Water-Sensitive Paper (WSP) diagnostic samples as feedback inputs to evaluate coverage density and automatically tune spray configurations to meet targeted chemical application thresholds for specific crop treatment regimes.

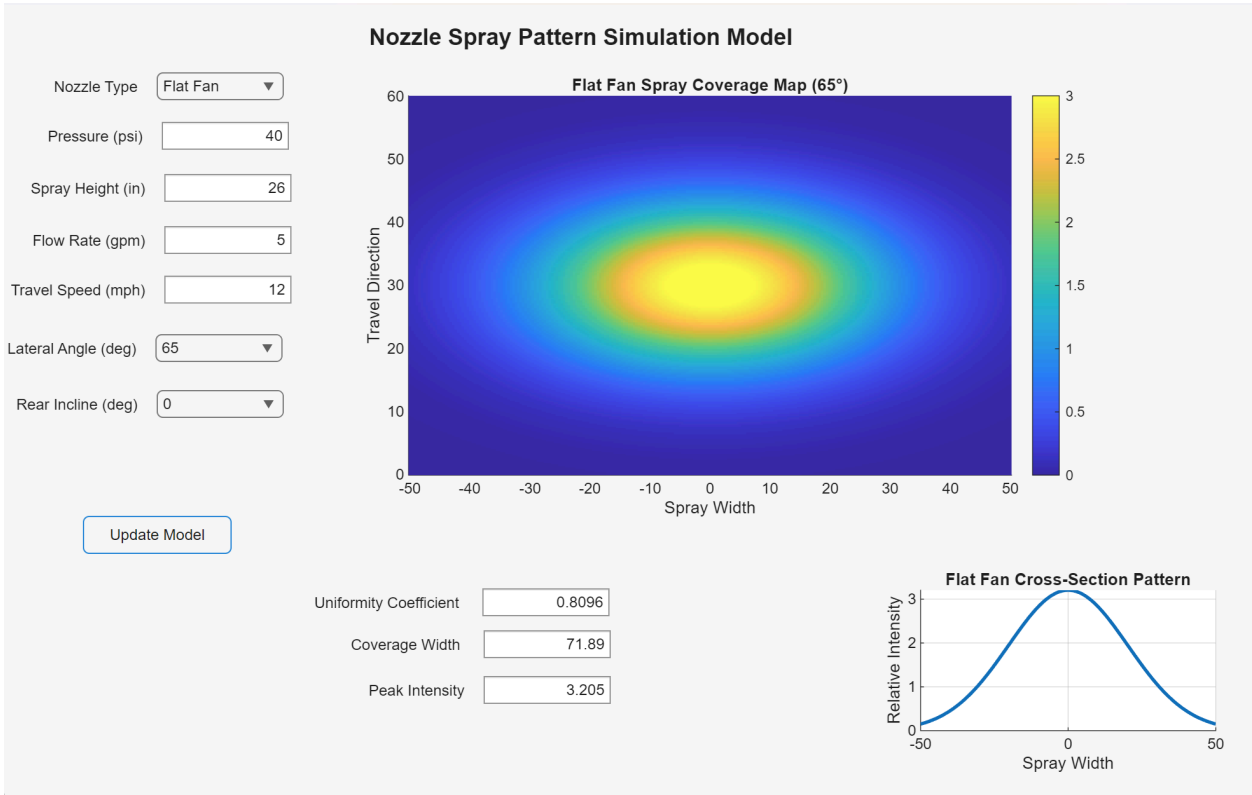
Examples:



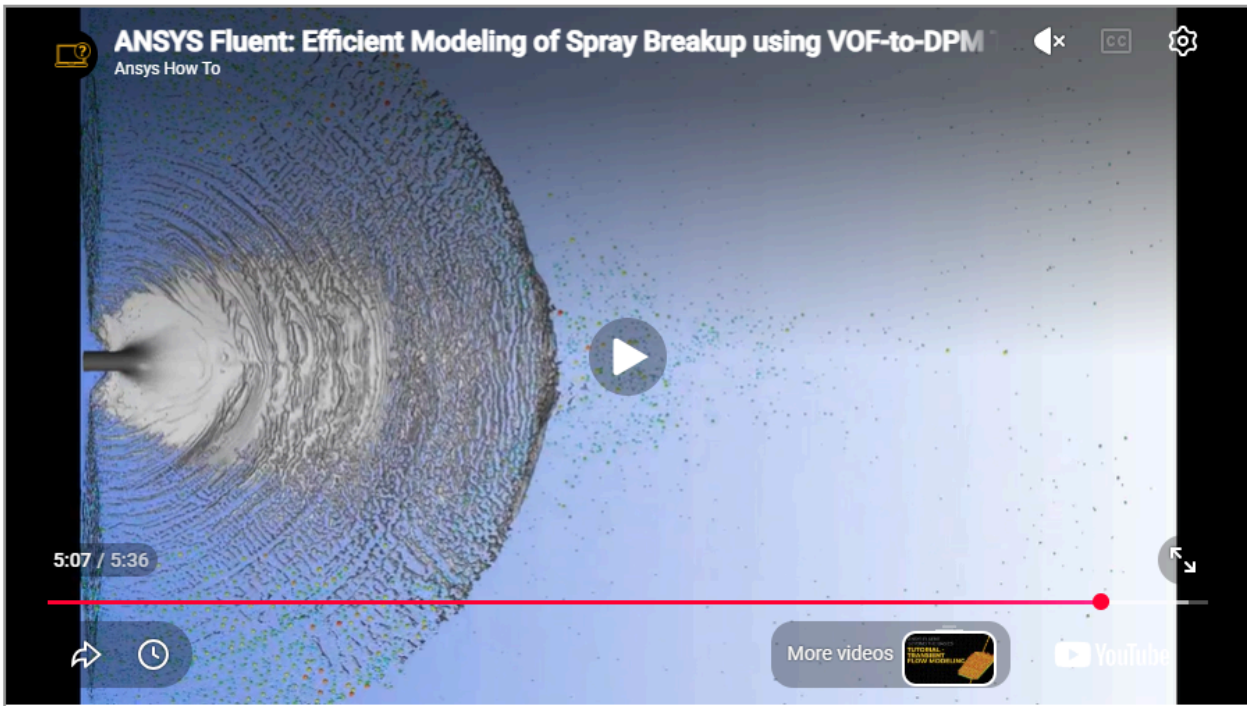
[Pesticide Spray Efficiency Simulation]



Spray Simulation (Redacted)



Spray Pattern Sensitivity Analysis, Emery Brewer 2026


ANSYS Fluent: Efficient Modeling of Spray Breakup using VOF-to-DPM
Ansys How To
5:07 / 5:36
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